Retail Profitability Analysis: Where Discounts Destroy Profit

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Introduction

A retailer can have strong sales and still lose money on a large share of its orders. This project looks at order data from a global superstore to find where profit is lost, what is behind it, and what the business could change.

Abstract

I analysed 51,290 order lines from the Global Superstore dataset (2011 to 2014). After cleaning the data, I used SQL in Google BigQuery to measure profit by sub-category and by discount level, and built a Power BI dashboard to present the results. The store earned 12.64M in sales and 1.47M in profit, a margin of 11.6%. However, 24.5% of order lines lost money. No order sold at full price lost money, while about 90% of orders with discounts above 20% did. Those orders lost about 815K, equal to 55% of total profit. I recommend capping discounts at 20%.

Tools Used

Python (Pandas) in Google Colab, Google BigQuery (SQL), Power BI Desktop, GitHub.

Steps Involved

1. **Data cleaning.** Checked for nulls and duplicates (no duplicates found), dropped the postal code column because 80.5% of it was empty, trimmed text and converted dates, and checked for impossible values. Added columns for profit margin, shipping days, year, month and season.
2. **Loading and preparation.** Loaded the cleaned file into BigQuery and created a view that groups discounts into four bands: none, 1-20%, 21-40% and above 40%.
3. **SQL analysis.** Calculated profit and margin by sub-category, margin by discount band for the whole store and for the Tables sub-category, and the share of order lines that lose money.
4. **Dashboard.** Built a Power BI dashboard (Import mode) with KPI cards, charts and filters for market, category, season and year.

Key Findings

Discount band	Order lines	Profit	Margin	Lines losing money
None	29,009	+1.77M	25.3%	0%
1-20%	10,953	+0.51M	13.7%	21.6%
21-40%	4,367	**-0.19M**	**-17.3%**	76.2%
Above 40%	6,961	**-0.63M**	**-74.1%**	98.4%

- **Losses start above a 20% discount.** Lines above that level are 22% of all lines but account for 81% of all losing lines.
- **Tables is the only sub-category that loses money overall** (-64K, average discount 29.1%). It earned +68K up to a 20% discount, then lost 133K above it.
- **Sales grew, profit did not keep up.** Monthly sales rose steadily from 2011 to 2014, while monthly profit stayed low and almost flat.



Figure 1. Power BI dashboard: KPI cards, profit by sub-category, margin by discount band and monthly sales and profit (2011 to 2014).

Conclusion

The store's problem is not low sales but unprofitable sales, and the data points clearly at deep discounts. I recommend capping discounts at 20% with manager approval above that, stopping deals above 40% first, and reviewing the pricing of Tables. The roughly 815K lost on heavily discounted lines is an upper bound on what a cap could recover, because some of those sales would not happen at lower discounts. The data shows that discounts and losses go together but cannot prove the discount alone caused each loss. The dataset also has no inventory data, so stock turnover could not be measured.